LIFTING A FACTORISATION OF A MONIC POLYNOMIAL AFTER AN ETALE BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lifting of an invertible element modulo an ideal after an etale base change inducing an isomorphism on the quotient. Lemma \ref{lemma-lift-invertible-element}, cited at proof lines 1915-1916. Makes the leading coefficient of the first factor the reduction of a unit.
- Lifting of a factorisation of a monic polynomial into monic factors. Lemma \ref{lemma-lift-factorization-monic}, cited at proof lines 1922-1923. Gives the factorisation once both factors are monic.

Source reference. Original lemma 07M1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9711–9711. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07LX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9707–9707. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0ALH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13370–13370. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a ring, let $I \subset A$ be an ideal. Let $f \in A[x]$ be a monic polynomial. Let $\overline{f} = \overline{g}\overline{h}$ be a factorization of f in $A/I[x]$ and assume*

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(1) *the leading coefficient of* $\overline{g}$ *is an invertible element of* A/I*, and*
(2) $\overline{g}$*,* $\overline{h}$ *generate the unit ideal in* $A/I[x]$*.*

Then there exists an étale ring map $A \to A'$ *which induces an isomorphism* $A/I \to A'/IA'$ *and a factorization* $f = g'h'$ *in* $A'[x]$ *lifting the given factorization over* A/I*.*

Proof. Applying Lemma 07LX we may assume that the leading coefficient of $\overline{g}$ is the reduction of an invertible element $u \in A$. Then we may replace $\overline{g}$ by $\overline{u}^{-1}\overline{g}$ and $\overline{h}$ by $\overline{u}\overline{h}$. Thus we may assume that $\overline{g}$ is monic. Since f is monic we conclude that $\overline{h}$ is monic too. In this case the result follows from Lemma 0ALH. □